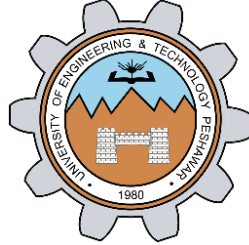


Normalized Relational Schema

LAB # 08



Spring 2024

CSE-403L Database Management Systems Lab

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Class Section: **C**

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Date:

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Objectives:

This lab aims at the understanding of:

- Normalized Relational Schema
- Relational Schema
- 3NF Normalized Relations

Task 8.1:

Submit the Complete Relational/Logical Schema of your DBMS Project including:

1. Relational Schema: Transformation of ERD and EERD into Relations
2. Normalization: 3NF of Relational Schema

Converted Relational Schema:

First Normal Form (1NF)

The current structure is in 1NF as all attributes contain atomic values.

Second Normal Form (2NF)

All tables satisfy 2NF as all non-key attributes depend on the entire primary key.

Third Normal Form (3NF)

The hall_id in Manager Table should not be in the Manager table as it creates a transitive dependency (hall_id depends on manager_id, and attributes in the Hall table depend on hall_id). Instead, we create a separate Manages table.

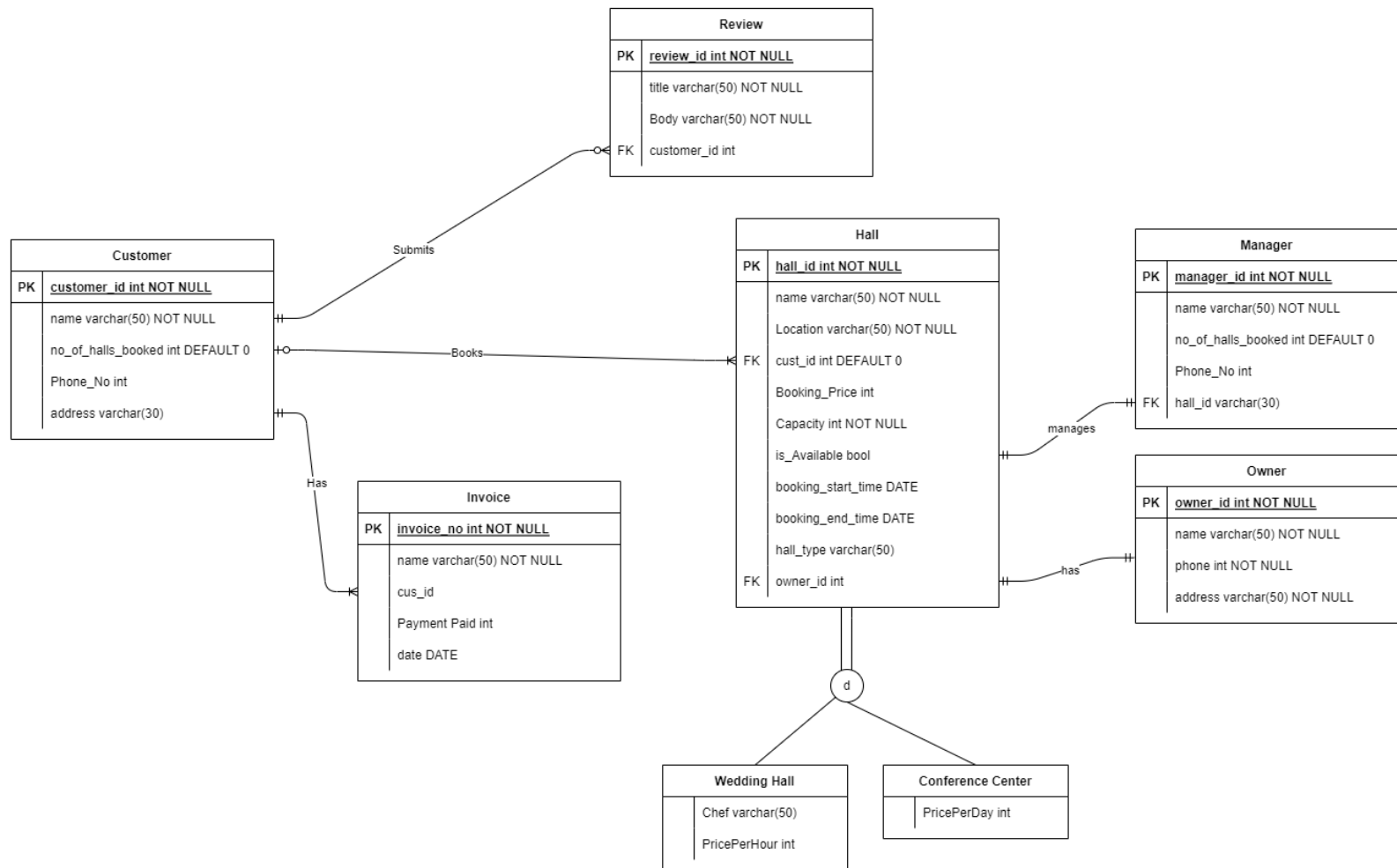


Figure 1-1, Unnormalized Relational Schema of Hall Booking System

Normalized Relational Schema:

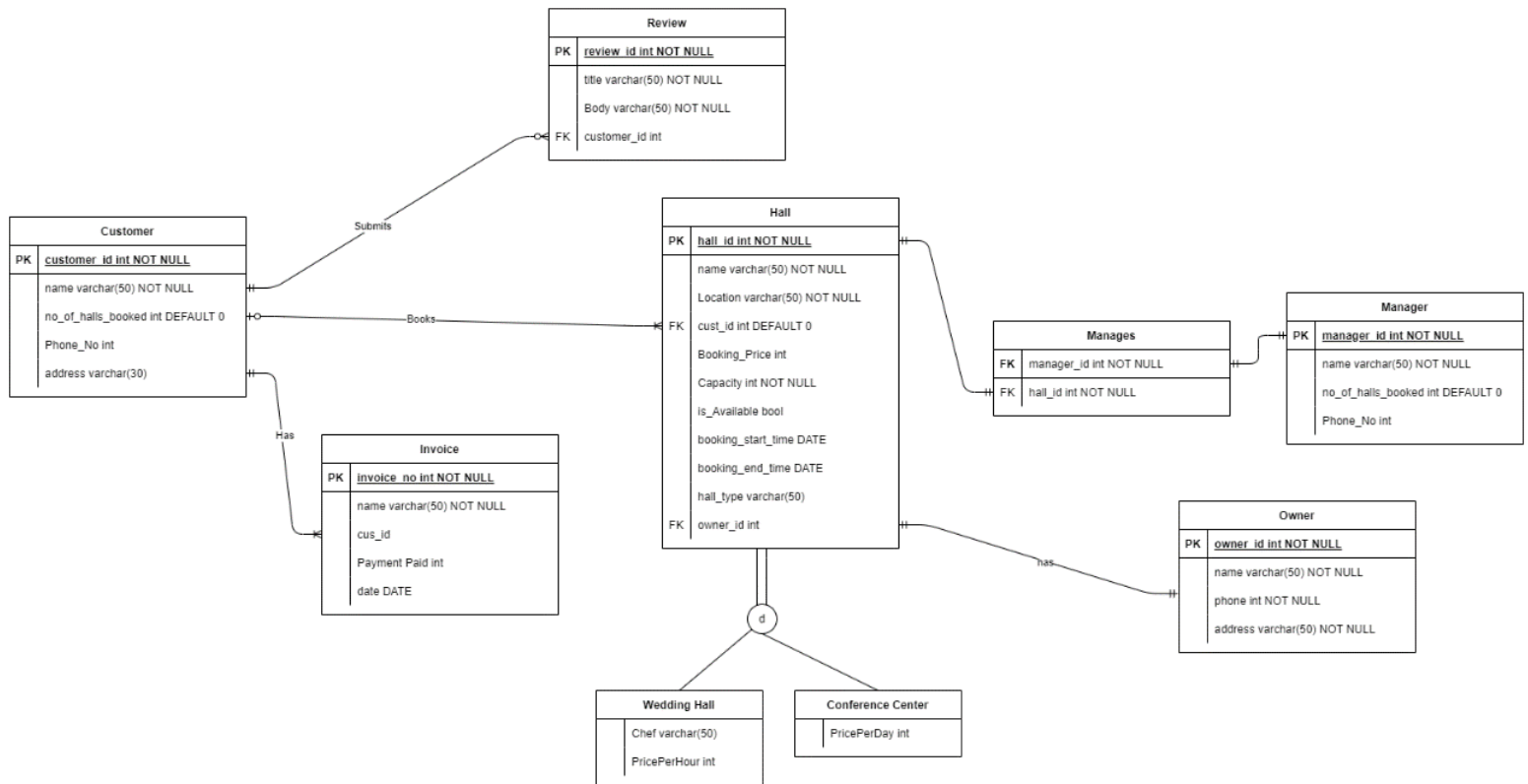


Figure 1-2, Normalized Relational Schema of Hall Booking System